

Section 3: What to pay attention to: lossy compression error and metrics

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How lossy compression errors affect scientific data?

- Users decide to use lossy compression algorithms for their data **only** if the compressed data result produces the same scientific outcomes as non-compressed data.

- Ideally, one could formalize mathematically the compression errors and propagate them into the simulation/analysis equations.

- In reality, it is very difficult. Even round-off error propagation is not well understood.

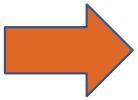
- We lack a profound understanding of the lossy compression error effects (result distortions, derived quantities distortions, convergence slow down) on different types of simulations, AI execution and experiments.

→ We need to build new tools and potentially new metrics to quantify lossy compression effects;

Assessment Levels & Mitigation of Adversarial Effects

Use-cases	Examples	Potential Adverse Effects
Visualization	Climate simulation	Visual quality, PSNR and SSIM alteration, creation of artifacts
Reducing data stream intensity	LCLS/APS[1] X-ray data christalography	No/wrong atom detection
Reducing footprint on storage	HACC[1]	Alteration of halo mass distribution and power spectrum
Accelerating check-point/restart	NWChem	No/slow/wrong convergence
Reducing footprint in memory	Quantum circuit simulation	Lower simulation accuracy
Accelerating execution	GAMESS[1]	Incorrect simulation results
AI training accelerating	CANDLE NT3	Classification errors
Deep learning model reduction	IOT[1]/sensors	Incorrect DNN[6] classification/regression results

8 use-cases



We need to:
-detect adverse effects
-understand the reasons behind adverse effects
-mitigate these effects

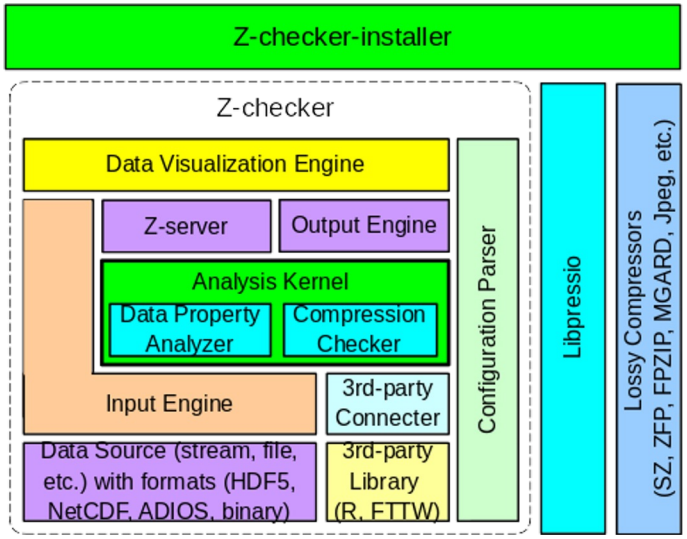
Glossary: LCLS: Linac Coherent Light Source. APS: Advanced Photon Source. HACC: Hardware/Hybrid Accelerated Cosmology Code. CESM: Community Earth Science Model. ATM: Atmospheric Model. GAMESS: General Atomic and Molecular Electronic Structure System. IOT: Internet of Things. DNN: Deep Neural Network

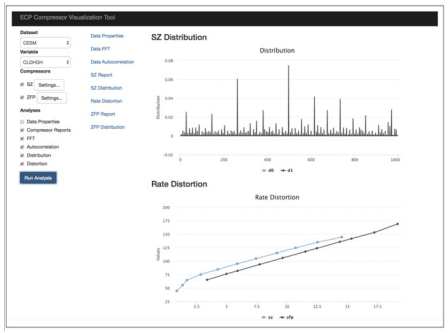
Assessing/Understanding Lossy Compression Impacts

When mathematical formulation is not possible, empirical assessment is necessary.

3 assessment levels to **detect** and **understand** adverse effects: different trade-offs between the quality and the practicality of the assessment.

- **L1**: Visualization (lossy compression comes from the vis. community). Vis. is a popular way to assess distortion.
- **L2**: Quantitative assessment of lossy compression errors.
  ANL Z-checker: first tool for off-line and on-line quantitative analysis of the compression error with >30 metrics
- **L3**: Quantitative assessment of the impact of lossy compression errors on users' analytics. Done manually until recently: LANL Foresight: first tool for L3 assessment.





Z-checker Interactive Error analysis Interface

Assessment Levels & Mitigation of Adversarial Effects

Use-cases	Examples	Potential Adverse Effects	Assessment Levels
Visualization	Climate simulation	Visual quality, PSNR and SSIM alteration, creation of artifacts	L1, L2
Reducing data stream intensity	LCLS/APS[1] X-ray data christalography	No/wrong atom detection	L1, L2, L3
Reducing footprint on storage	HACC[1]	Alteration of halo mass distribution and power spectrum	L2, L3
Accelerating check-point/restart	NWChem	No/slow/wrong convergence	L3
Reducing footprint in memory	Quantum circuit simulation	Lower simulation accuracy	L3 L2
Accelerating execution	GAMESS[1]	Incorrect simulation results	L3 L2
AI training accelerating	CANDLE NT3	Classification errors	L3
Deep learning model reduction	IOT[1]/sensors	Incorrect DNN[6] classification/regression results	L3

8 use-cases

In all these cases:

-L2 analysis performed with Z-checker

-L3 analysis performed manually (except Foresight was used for HACC)

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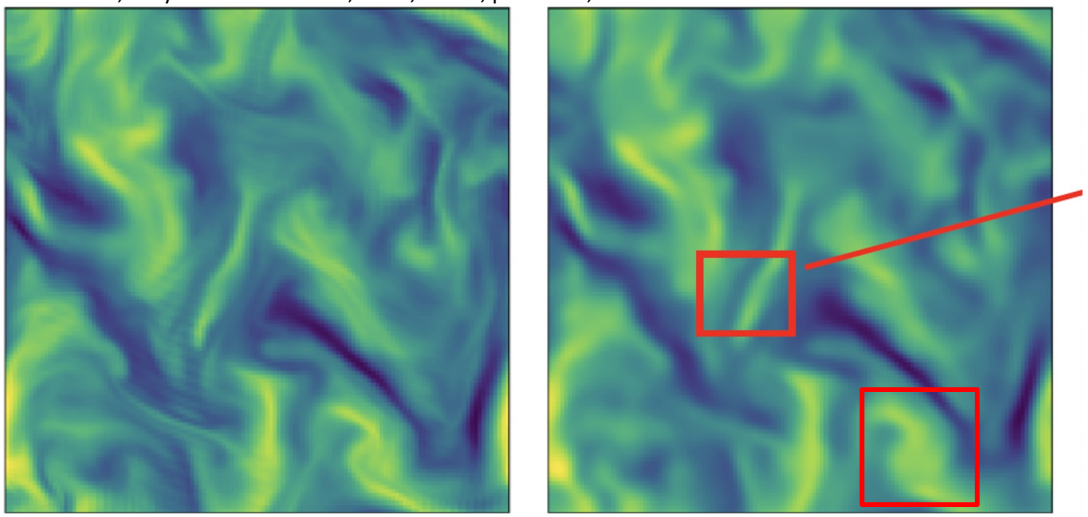
Level 1: Visualization: Data: looking for artifacts

Artifacts on data itself

Turbulence dataset compressed with Autoencoder

Too much smoothing in this case

A. Glaws, R. King, and M. Sprague, "Deep learning for in situ data compression of large turbulent flow simulations," Physical Review Fluids, vol. 5, no. 11, p. 114602, 2020

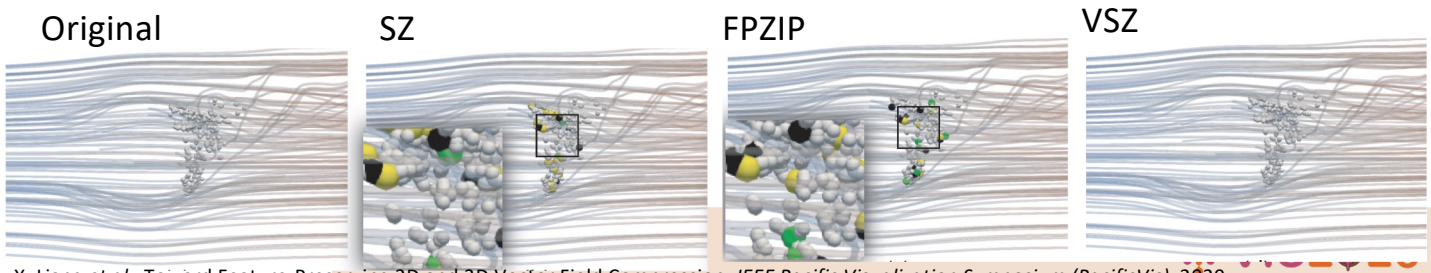


Original data AE reconstructed data (64:1)

Artifacts on derived quantities

Critical points of vector fields (important for flow visualization):

Added, moved or missing critical points in this case



X. Liang *et al.*, Toward Feature-Preserving 2D and 3D Vector Field Compression, *IEEE Pacific Visualization Symposium (PacificVis)*, 2020

Level 2: Quantitative assessment of lossy compression errors

- **Error Metrics (amount of error)**
  - Respect of error bounds
  - RMSE
  - PNSR
  - Rate distortion

Point wise metrics

Statistical metrics

- **Advanced Error Metrics (nature of the error)**
  - Error distribution
  - Pearson correlation of the initial and decompressed data
  - Autocorrelation of the compression error
  - Correlation between the initial data and the error
  - Power spectrum alteration
  - Structural Similarity Index (SSIM)
  - Preservation of derivatives

Level 2: PSNR: Peak Signal to Noise Ratio

Based on the ratio between the data range and the root mean square error

$$X = \{x_1, x_2, ..., x_N\} \Rightarrow \text{Compression / Decompression} \Rightarrow \tilde{X} = \{\tilde{x_1}, \tilde{x_2}, ..., \tilde{x_N}\}$$

Value range: $R_X = x_{max} - x_{min}$

$e_{absi} = x_i - \tilde{x_i}$

Root mean square error:

$$rmse = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(e_{abs_i})^2}$$

Peak signal-to-noise ratio:

$$psnr = 20 \cdot log_{10}\left(\frac{R_X}{rmse}\right)$$

Why log10: Because of the dynamic range of datasets (often orders of magnitude)
Why 20: because of 10 (Deci-Bel) x 2 (Rx/rmse is Rx^2/(rmse^2) before log rule application)

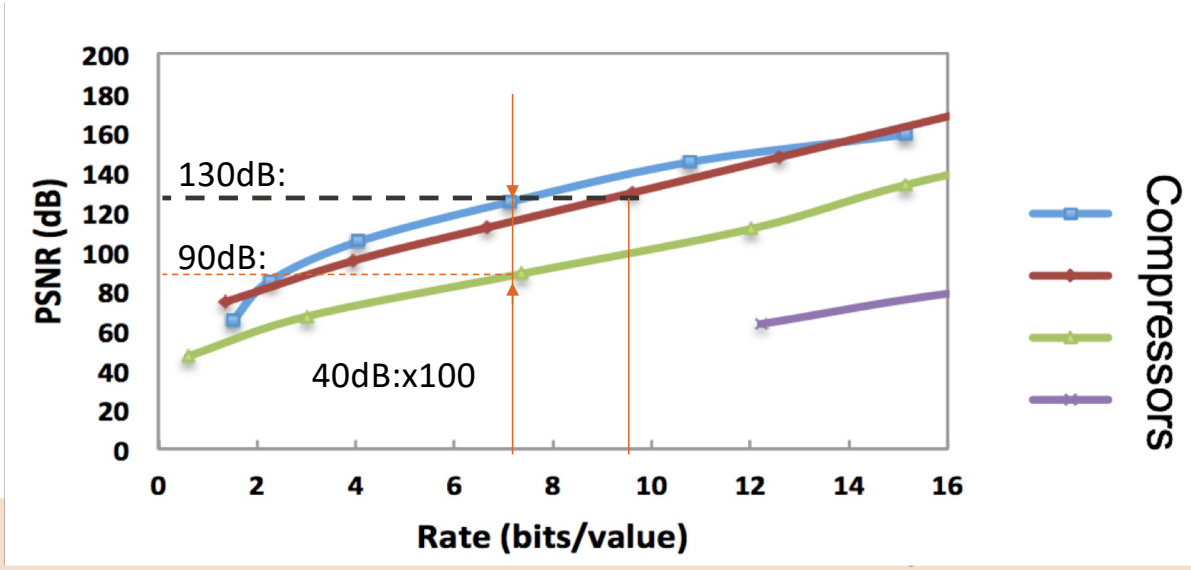
dB	Amplitude ratio
-100 dB	10^{-5}
-50 dB	0.00316
-40 dB	0.010
-30 dB	0.032
-20 dB	0.1
-10 dB	0.316
-6 dB	0.501
-3 dB	0.708
-2 dB	0.794
-1 dB	0.891
0 dB	1
1 dB	1.122
2 dB	1.259
3 dB	1.413
6 dB	2 ≈ 1.995
10 dB	3.162
20 dB	10
30 dB	31.623
40 dB	100
50 dB	316.228
100 dB	10^5

Level 2: Rate-Distortion

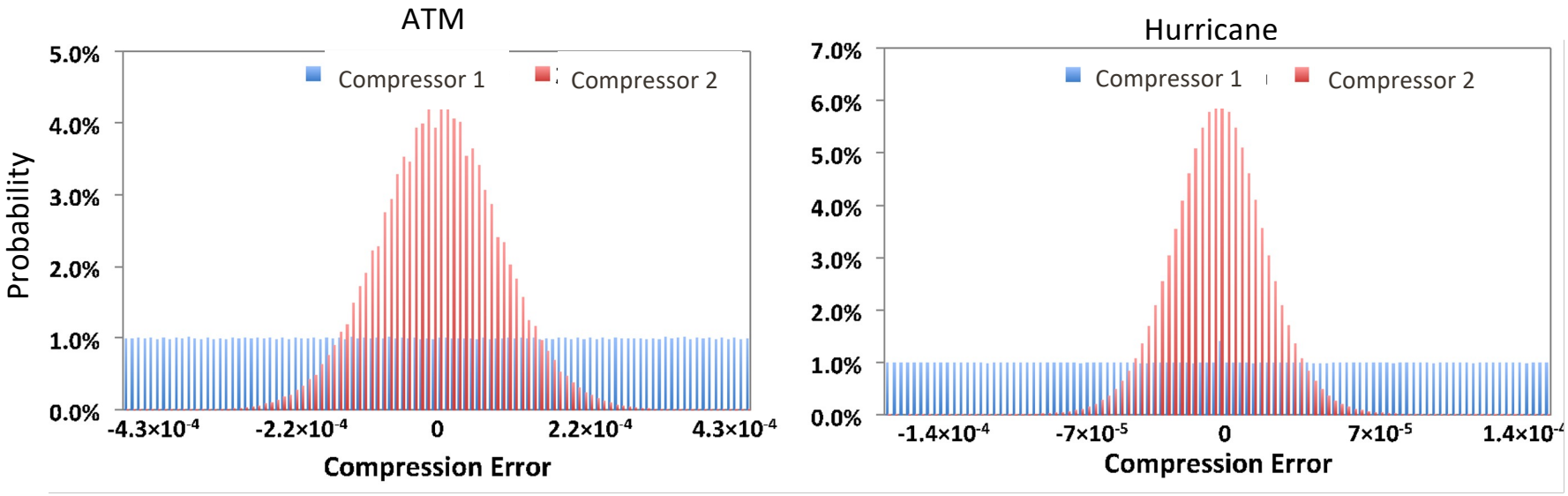
Metrics that comes from the information theory: determine the minimal number of bits per symbol, as measured by the rate *R*, that should be communicated over a channel, so that the source (input signal) can be approximately reconstructed at the receiver (output signal) without exceeding a given distortion *D (PSNR)*

In our case: Rate = #bits per floating point value in the compressed data sets

Compress the data set for different compression ratios (by fixing the rate or by fixing error bounds), measure PSNR and plot a graph with the measurements



Level 2: Distribution of Compression Error



Uniform: SZ 1.4

Gaussian: ZFP, SZ 2.0

Distribution plot informs about the nature of the noise that the lossy compressor adds to the dataset. It also shows how the **errors are distributed around 0**. SZ and ZFP provide **symmetric errors**: Good. Preferred distribution (Uniform or Gaussian) is application dependent.

Level2: Autocorrelation of the compression error

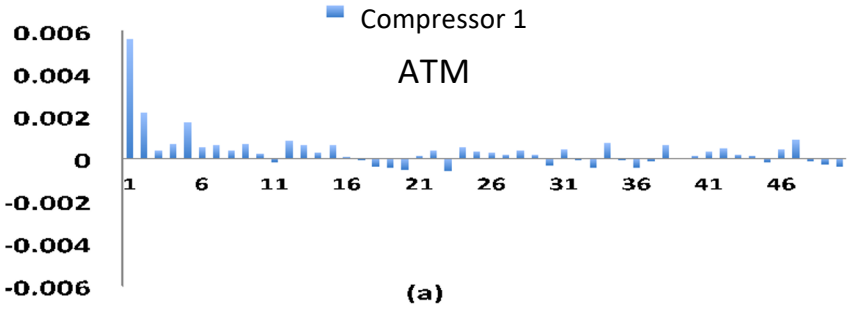
Autocorrelation of a random process:

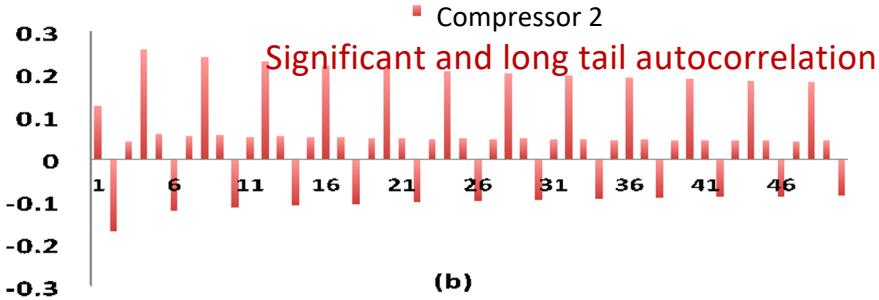
$$R(\tau) = \frac{\mathrm{E}[(X_t - \mu)(X_{t+\tau} - \mu)]}{\sigma^2}$$

expected value · mean · Autocovariance · Autocorrelation for lag · Variance

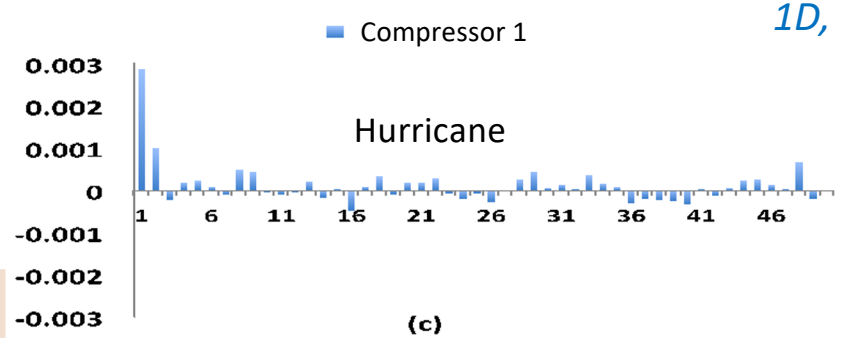
is the Pearson correlation (still between 1 and -1) between values of the process at different times (or lags): T and T+t

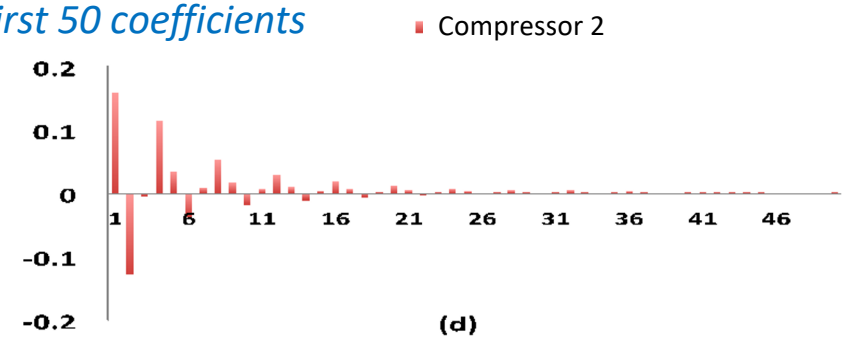
We want as little (0) autocorrelation of the compression error as possible (amplitude and tail)





1D, First 50 coefficients





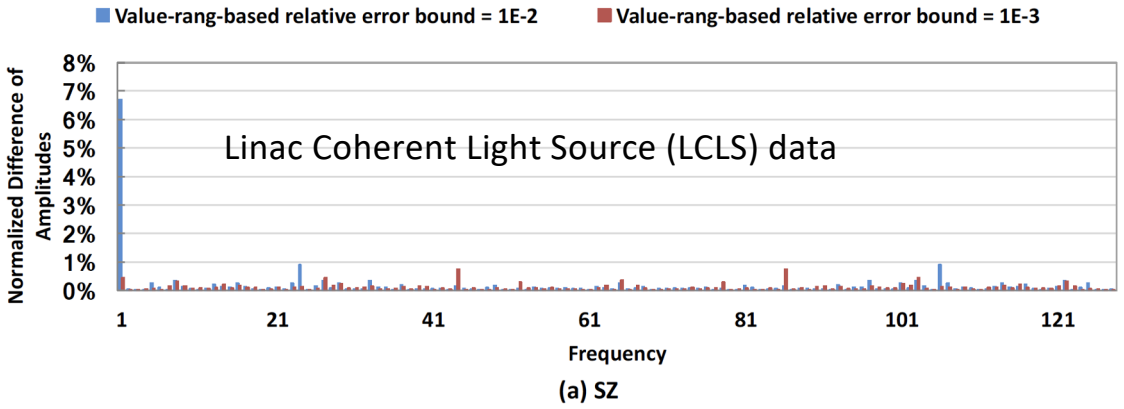
Level 2: Frequency spectrum alteration

We want to know how the compression affects the different components of the signal (frequency domain)

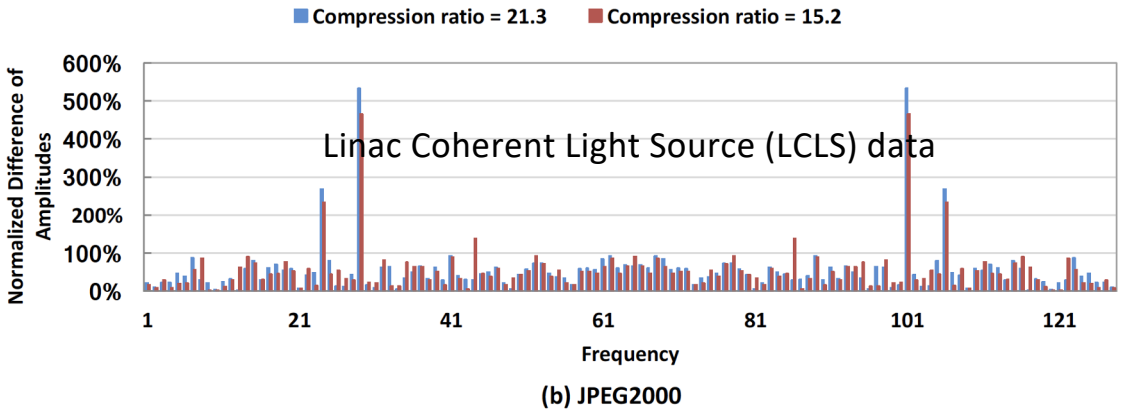
Compute the Fourier Spectrums for
- the initial data set
- the decompressed data set

Plot the amplitude difference for each frequency (normalized from the initial data set)

In general, we want to preserve as much as possible the spectral properties of the signal



(a) SZ



(b) JPEG2000

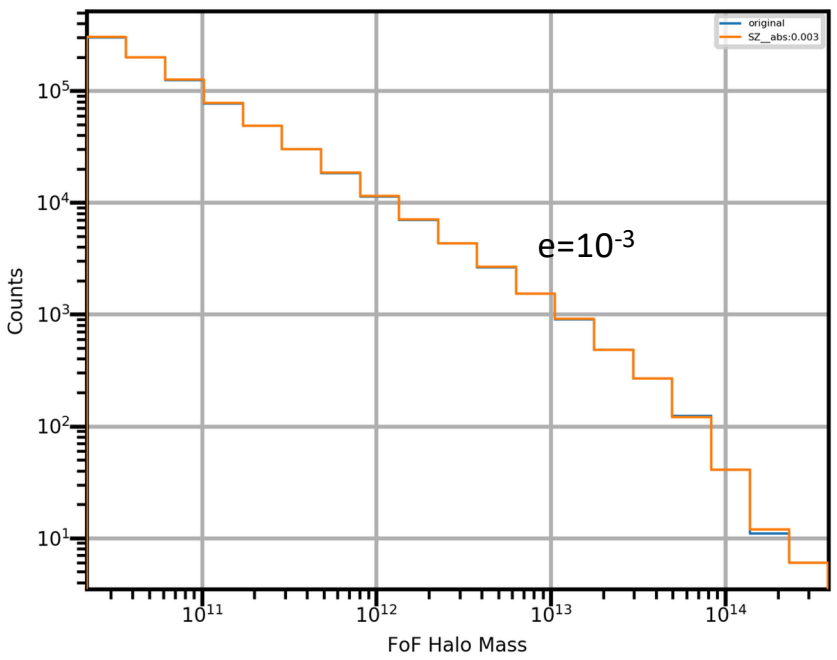
Level 3: LANL Foresight on HACC Data

Graphs generated by **LANL Foresight**. Foresight is a Level 3 compressor error assessment tool
https://github.com/lanl/VizAly-Foresight

Results validation

3kpc absolute error bound
(particle position)

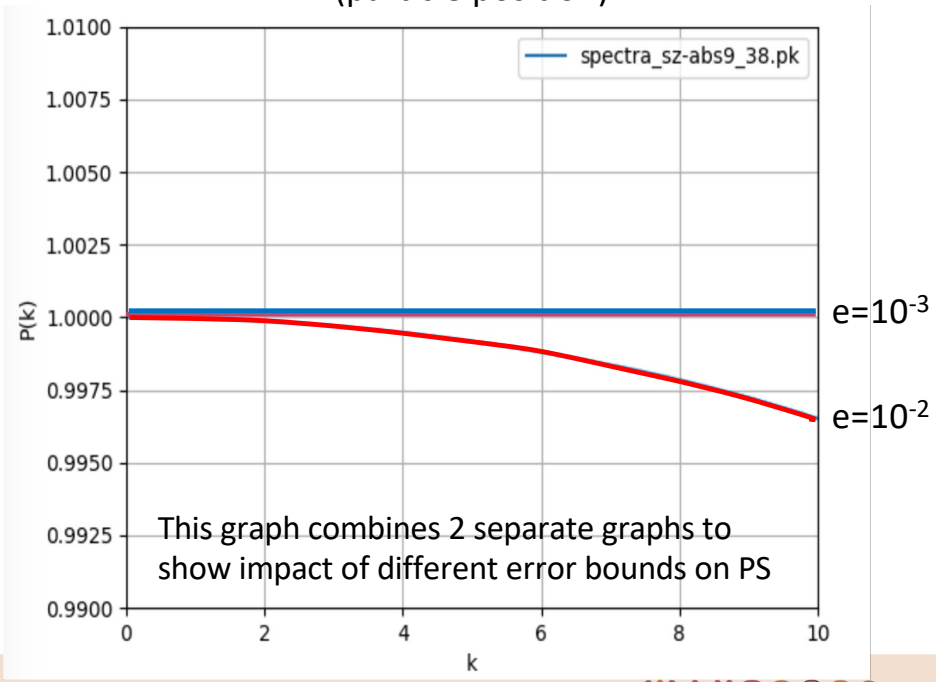
Original:
SZ ($e=10^{-3}$):
SZ ($e=10^{-2}$):



Friends of Friends halo mass distribution

Results validation

3kpc absolute error bound
(particle position)



Power Spectrum

Assessment Levels & Mitigation of Adversarial Effects

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Reducing footprint on storage	HACC[1]	Alteration of halo mass distribution and power spectrum	L2, L3	Improvement of compression algorithm, compression parameter tuning
Accelerating checkpoint/restart	NWChem	No/slow/wrong convergence	L3	Compression parameter tuning
Reducing footprint in memory	Quantum circuit simulation	Lower simulation accuracy	L3 L2	New compression algorithm
Accelerating execution	GAMESS[1]	Incorrect simulation results	L3 L2	New compression algorithm
AI training accelerating	CANDLE NT3	Classification errors	L3	Compression parameter tuning
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